OpenSourceRail v0.5

Ferrite PM maglev — primary design reader edition

Fixed magnetic modules · compact elevated guidebeams · Rust controls · local manufacture

Provisional digital studies. Complete vehicle, civil and software qualification remains open. Accepted physical releases: 0.

OpenSourceRail v0.5.2 development — fixed-zone ferrite maglev

docs/open-source-rail-overview.md

Fewer moving parts, local manufacture and compact elevated urban routes.



OpenSourceRail maglev station concept

Illustrative development concept, without physical or manufacturing acceptance.

Fixed linear synchronous motors launch and receive the train at stations. Selected longer legs add located midway acceleration; remaining guideway is unpowered; onboard ferrite PM-biased active support, guidance and passenger services remain independently powered. Removing main traction equipment from the train shifts electrical plant to stations. Support coils/drives, cooling, sensing, protection, doors, HVAC and emergency equipment remain necessary. Contactless running aims to reduce contact noise and wear; actual losses, acoustic performance and maintenance require measurement.

Body, civil works and local capability

Baghdad uses six 17 m bodies. The central tare target is 14.2 t/car, including a 5.2 t equipped structural GFRP body, 3 t ferrite, 30 kWh services storage and an existing 1.8 t growth reserve. Four passenger doors per car are the body takeoff. Counted frame families, supported panels, couplers and hardpoint candidates now have calculated dimensions and margins; frame/emergency-brake growth consumes this reserve. Joined-body strength, complete equipment, fatigue, CG and inertia remain unqualified. The50kWh wider-corridor reference is 14.4t tare/ 26.4t AW3 per car. Its pack increase is included once; the primary detail/emergency screens use that heavier mass alongside the retained30kWh case.

The current ordinary civil candidate is a 25 m, 0.80 × 2.50 m closed concrete box with 100 mm webs, 150 mm flanges and end infills. The supplied 1.80 m-depth case remains a separate comparison. Hollow columns/caps and alternative foundation families are compared by quantity. Necessary passive iron, anchors, bearings, connections and final survey remain in scope. Ground-specific selection, prestress/handling details, fixed/guided bearings and emergency load paths have numerical screens; physical structural and construction acceptance remain open.

The research target omits a continuous external walkway. Vertical spiral chutes are the preferred alternative under investigation, without walkway-removal approval. Safe landing, assisted evacuation, deployment, stopping/recovery and fire

response must work at every allowed stopping location.



Vertical spiral escape chute concept

Illustrative concept; no qualified railway device or all-stop evacuation claim.

The 99% elevated-mainline objective and straight tangents aim to reach constrained corridors with less surface occupation. Rights, buildings, utilities, access, wildlife, water, curvature and foundation feasibility govern each route.

Local precast/GFRP production, steel fabrication, cartridge assembly, station/booster winding cassettes, interiors, wiring, Rust software and integration can build domestic capability. Prioritise verified British manufacture when selecting UK machinery. A UK distributor or integrator does not establish manufacturing origin. Complete local content remains unknown until the bill and input origins exist.

Operating and cost boundaries

Urban passenger stops target at least 1.5 km spacing and about 2 km on average. Suburban stops target 4–5 km with fixed midway acceleration where required. Junctions transfer passengers between independent fixed lines, without train-routing switches. Retained close junctions remain spacing/access exceptions requiring a solution. Straight alignments are planning candidates subject to land, buildings, utilities, grades, access and evacuation.

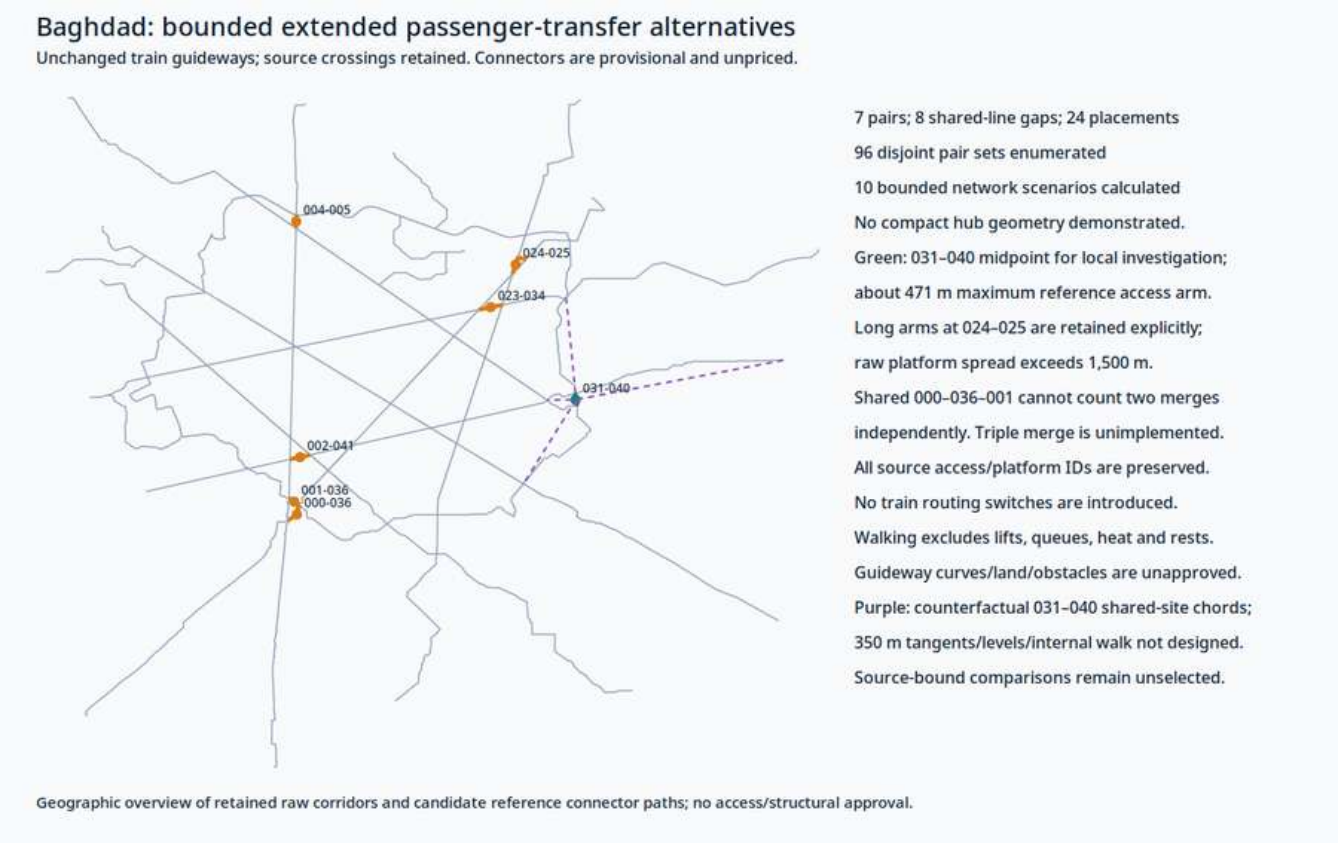
The 1 km/88 s/180 s station and moving packets remain component benchmarks. The supplied 90 s/190 s case is a separate authored comparison. Longer urban/suburban profiles have their own finite-zone, power, services, clearance and actual Rust fault tests. Capacity shortfalls and unimplemented sectioned protection stay visible; a timer does not prove a clear railway.

The retained benchmark uses 240 kW/car charger input at 90% assumed conversion efficiency. Longer cases compare their own pack, charger and dwell choices. Independently equipped rescue nodes, faster winches and 30/50/60 kWh packs have explicit rope/drum/portal/foundation, drive/power, mass and cost comparisons. A midpoint accelerator is not automatically a refuge, charging station or rescue site. Rope/termination, access, essential loads, physical hardware and real ground remain unqualified.

City subtotals separately count eight scopes: fleet, ordinary and terminal civil work, station electrical targets, a conditional footing increment, booster electrical plant, booster mount stock and independent rescue kits. Complete station/transfer hubs, grid reinforcement, land, factory capital, evacuation, qualification, finance and other unpriced work remain additional. Fewer passenger platforms do not by themselves establish a total saving.

Baghdad interchange investigation

The default Baghdad plan remains 273 proposed stops with 10 transfer-spacing conflicts. A separate interchange investigation compares extended walking links on unchanged guideways with coordinated shared-site hub concepts that change guideway geometry. No alternative is selected.



Retained Baghdad corridors, extended transfer links and unselected shared-site concept

Original boarding/access records remain traceable. The extended links retain reference walks and partial electrical target credits, without complete connector costs or ridership-weighted benefits. Shared-site concepts draw involved fixed lines through one location, but station tangents, curves, levels, platforms and actual internal walks remain undesigned;100m is only a compact-walk sensitivity. Lift resilience, peak transfer pulses, shade, resting places, evacuation, land and complete construction/maintenance costs must enter selection. Trains never switch between lines in these passenger-transfer proposals.

Inspect and reproduce the current work

Current digital capability	Scope
City planning	266 primary packages; located operating stops, access assignments, passenger-transfer spacing conflicts, boosters, rescue nodes, motion/headway/capacity screens and eight-part cost subtotals
Operating study	Separate urban/suburban profiles with finite station/booster zones, services/charging and electrical/protection fault cases
Interchange investigation	Unselected Baghdad extended-access variants and shared-site guideway concepts; default plan and open acceptance preserved
Manufactured details	Counted frame/panel/hardpoint/coupler/stator interfaces, prestress/handling and emergency load-path screens, foundation cost deltas and open qualification
Controls	Actual Rust support replay and integrated station/resource campaign with full-tail clearance and fault cases; one local controller and virtual detectors
Simulation and media	Retained 1 km six-car moving numerical run with repeated flexible spans and couplers, calculated-motion video, partial native detail CAD/images and visibility videos
Engineering acceptance	0 accepted physical releases; design, manufacture and operational gates remain open

Ring benchmarks and block-orientation seeds receive no railway lift or mass-saving credit. The moving run uses ideal independently allocated magnetic forces, rigid cars, linear beams and virtual sensing. Real force maps, shared-field phase, flexible bodies, lateral/soil/contact response, hardware timing and distributed controller handover remain open. The moving video uses calculated positions with declared amplification; other videos show final-position stock visibility, without collision or erection-strength acceptance.

Use `./osr build` to verify dependencies and rebuild stale implemented studies. `./osr design station`, `./osr design structure` and `./osr design protocol` reproduce their separate evidence. `./osr workbench` opens the review UI; `./osr book` builds the current reader edition. The scheduled runner refreshes implemented evaluators after source changes; it does not invent physical evidence.

Open the Workbench, review calculations, CAD and videos, read implementation and open work, inspect moving six-car evidence, inspect the manufactured dossier, inspect wider-stop corridor evidence, inspect booster and rescue quantities, compare Baghdad interchange alternatives, review v0.5.2 development notes, or download the reader book.

Published v0.5.1 remains preserved. This document describes v0.5.2 development on main and does not announce a new release.

OpenSourceRail v0.5 fixed-zone ferrite-maglev engineering

engineering/ferrite_maglev/README.md

The primary design uses fixed linear synchronous motors at stations for launch and receiving braking, plus located midway acceleration on longer legs. Guideway outside these finite powered zones remains unpowered. Onboard ferrite PM-biased active support/guidance and services storage remain independent. Structural GFRP sandwich bodies, repeated narrow concrete guidebeams and Rust control packages are the leading studies. Accepted engineering and physical releases: zero.

Run and inspect

```
./osr build
./osr design station
./osr design structure
./osr design protocol
./osr design corridor
./osr design booster-kit
./osr design iterate
./osr design check
./osr design verify
./osr city baghdad
./osr sim baghdad --cycles 2
./osr workbench
./osr test
./osr book
cargo test --locked -p osr-maglev-control --all-targets
```

Current report · Workbench · city directory · architecture.

`./osr capabilities` publishes executable sources and scope. `./osr catalogue plan --city baghdad` resolves a selected regeneration, and `./osr catalogue run --city baghdad` rebuilds that city. `./osr readiness` reports source and acceptance gates. `./osr design release-check` remains held while engineering acceptance is open.

The dependency runner verifies implemented native, numerical, control, city and publication prerequisites. An interprocess lock excludes concurrent timer/manual builds. Progress is retained under `build/maglev-workflow/latest.json`. The runner rebuilds stale evidence and holds dependent work after failures; missing evaluators and physical measurements remain additional engineering work.

Current inputs and design records

Concern	Source
Architecture	Infrastructure policy
Vehicle/civil configuration	Configuration, reader, primary inputs
Wider-spacing operating corridors	Urban/suburban model, actual trajectories, Rust replay and limits
Booster and independent recovery hardware	Kit model, quantities and alternatives, native kit
Station motion and electrical plant	Operating model, published trajectory/overlap/power study
GFRP, magnet orientations and civil production	Structure model, study inputs, published quantities
Requirements and allocation	Requirements, systems register
Costs and manufacture	Process model, planning rates
Magnetic/native geometry	Geometry, field controls, native exporter, station-native packet
Station reservation protocol	Offline protocol evidence, persistent Rust interlocking ownership
Moving six-car twin	Numerical model, run/controls/video
Manufactured details	Dossier, specifications/margins/obligations, native solids
Integrated station/control campaign	Executable campaign, command/input/output evidence
City geography/service	Each city's `city.json`; compiler, operating-plan generator
Baghdad interchange alternatives	Investigation model, unselected access/shared-site comparisons, map
Onboard support control	Rust package, offline host profile

Baghdad uses six 17 m bodies with 0.5 m assumed coupler gaps. The retained 30 kWh tare target is 14.2 t/car, including a 5.2 t equipped GFRP body, 3 t ferrite and an existing 1.8 t growth reserve. The 50 kWh operating reference is 14.4 t tare / 26.4 t AW3 per car (158.4 t six-car AW3); the pack increase is included once in mass. Four passenger doors per car are the body takeoff. The manufactured dossier uses that heavier case for hardpoints, passive supports, civil demands and exact brake energy/thermal quantities, alongside a named 30 kWh benchmark. Its counted primary-frame overrun and heavier brake proposal consume growth reserve; battery mass is not charged a second time. Unweighed hardware, joined-body behavior, fatigue, CG and inertia remain open.

The primary ordinary box uses 25 m spans, 0.80 m width, 100 mm webs, 150 mm flanges and end infills. Selected depth, stiffness and mass derive from current configuration and retained numerical evidence; the supplied 1.80 m case is a comparison. The detail dossier adds prestress/handling alternatives, one fixed and one guided bearing per span, longitudinal brake-load distribution and hollow-support/footing sensitivities. A larger footing is a conditional priced proposal under declared soil/load assumptions, not a geotechnical selection.

Implemented checks and open boundaries

The separate Baghdad interchange investigation leaves the default273 stops and10 spacing conflicts unchanged. It compares24 extended-transfer placements on unchanged guideways and bounded disjoint network scenarios, retaining rejected extent controls and overlapping-pair constraints. Seven coordinated shared-site concepts instead draw changed guideway chords to a common location. None is selected or physically qualified. Shared-site tangents, curves, levels, platform/bridge structures and actual internal walks remain undesigned; their100m walk is a sensitivity assumption. Extended-link quantities and electrical target credits exclude complete connectors, station civil and passenger-time economics. No equal-capacity savings or compact-hub proof follows from grouping IDs.

Capability	Implemented scope	Evidence still required
Station trajectory	Jerk-limited launch, zero-thrust coast and receiving brake; energy closure and assumed magnetic-band overlap	Measured drag/magnetic losses, actual grades/wind, useful thrust, voltage/current/thermal limits and robust arrival margins
Station electrical costs	Fixed-group/section alternatives and separate power/charging target-rate rows	Configured equipment, supplier quotations, grid capacity, real duty and complete station scope
Body/civil quantities	GFRP panel mass cases, interface ownership, beam/end-infill quantities, support/foundation screens and target costs	Orthotropic structural analysis, member/joint/prestress/rebar detailing, soils, tolerances and construction-stage acceptance
Magnet orientations	Point-dipole ring reference and finite-block orientation inventories	Nonlinear finite-geometry steel/coil force/loss/demagnetization maps and measured shared-field behavior; no ring-gain credit
Native geometry	Partial panel/interface/civil/station definitions, export/reopening checks, retained images and visibility sequences	Complete production geometry, joints, equipment, drawings and qualified manufacturing tolerances
Rust support control	Module allocation, virtual replay, protection interfaces and support-preserving stop obligations	Actual magnetic/sensor plant, independent protection, physical I/O and measured timing
Launch ownership	Offline corridor/receiving-berth reservation, committed occupancy and full-clearance protocol checks	Authenticated complete-consist sensing, real receiving readiness, independent stopping and qualified system integration
Moving six-car replay	Six independent rigid-car states, actual Rust support allocation, repeated vertical flexible spans, longitudinal spring/damper chain, numerical controls and rejected comparisons	Measured magnetic maps, shared-field phase, normal/thrust coupling, flexible bodies, lateral/torsional/joint/soil/contact response and physical timing
Manufactured details	Counted frame stock, supported panels, joints, couplers, stator/bearing interfaces and handling/emergency load paths	Orthotropic laminates, joined load transfer, tolerances, fatigue/crash, calibrated limits and independent qualification
Integrated campaign	Actual Rust resource ownership and station control over incoming trip, dwell, next departure and negative cases	Authenticated sensors/plant, distributed handover, real receiving readiness and independent physical protection
Adaptive design	Previous-result-guided vehicle/beam/gain proposals, ancestry, Pareto observations and separate confirmation	Completed parts and qualified evaluators/plant before manufacturing or operating selection
Factory and city delivery	Conditional two-shift bottleneck capacities, duty/fleet, route quantities and process needs	Measured cure/crew/logistics rates, surveys, route recovery, complete economics and accepted instances

The adaptive search, configured in iteration.toml, has explicit evaluation/stagnation budgets and exact-source restart receipts. Failed confirmation disqualifies the exact revision. Published history retains proposals, ancestry, traces and recommendation. Mass/inertia allocations and preference weights remain research assumptions. An analytical recommendation does not automatically change acceptance criteria or reference configuration.

Operation, manufacture and costs

The primary operating strategy targets urban stops at least 1.5 km apart and about 2 km on average, with 4–5 km suburban legs and located midway boosters where required. Fixed-line junctions are passenger transfers, without physical train-routing switches. The retained 1 km/88s/180s component benchmark is distinct from those longer operational profiles. The source-bound station-powered packet preserves the supplied 90 s/190 s benchmark. Actual city intervals and headways derive from bounded motion, overlap and full-tail-clearance searches. Stops are placed at passenger transfers and appropriate intervals, with retained access assignments. Nearby mandatory transfers that violate minimum spacing are reported explicitly; straight-chord alignment candidates preserve crossing nodes but do not approve clearances, bends or land. Terminal-loop proposals remain separately unqualified. Original access counts, operating stops, selected capacity and any shortfall are explicit.

A committed train in a coast or booster corridor keeps protected resources until positive full-consist clearance; timeout cannot revoke physical occupancy. The actual Rust campaign joins this ownership to station control for two consecutive legs, and retains receiver/sensor/drive/resistor/stale/phase/uncommitted fault cases. Virtual detectors and one local controller do not establish distributed handover. Independent friction-brake and winch recovery proposals now have force, energy, reserve

and civil load-path calculations; calibrated friction/thermal hardware, access, curved recovery and physical stopping/holding remain unqualified.

At 60 kW/car hot services, the authored 90 s trip plus 15 s disconnected dwell consumes 10.5 kWh/train. Replacing this in 45 s takes 840 kW plus 360 kW direct services, or 1.20 MW/train before losses. The retained benchmark uses240 kW/car input and90% assumed conversion. Longer operating cases evaluate their own charger, battery, dwell and repeated services-energy requirements; those ratings are not inherited from the1km benchmark. The failing 200 kW input comparison is retained. Charger rating, taper, ageing, thermal duty and grid capability remain open.

City priced-scope subtotals separately label fleet, ordinary elevated guideway, terminal-loop civil work, station electrical targets, a conditional larger-footing increment, located booster plant and independent rescue kits. The footing delta accounts for concrete, reinforcement and earthworks under an unqualified soil/load proposal. The authored USD 4.255 million versus USD 3.313 million ordinary route-km comparison remains a narrow historical target-rate comparison, not the price of the new complete operating plan. No supplier quote or complete installed programme price exists. Stations, land, depot, external grid, evacuation, factories, acceptance, recovery, finance and other unpriced scope remain additional.

The booster kit counts2.5 m face cassettes,12.5 m electrical sections, mounts, bolts, cabinet pads and actual spatial reactions at fixed/guided bearings. Equivalent I^2R duty is integrated without inventing winding temperatures. Complete winding/core/cooling masses and nonlinear magnetic/electrical/thermal maps remain open. The partial native scene contains defined mount/drum/portal/anchor stock, without claiming a complete motor or assembled interference check.

Recovery alternatives compare ≤1km accessible node spacing with midpoint-only nodes,30/50/60 kWh packs and0.5/1 m/s towing. The5 km/300 s hot study retains the failing30 kWh midpoint-only slow-winch case. Standalone rope/drum/portal/foundation, rated drive/power allowances and QA/access costs are additional; co-location with an existing station, pier or booster earns no automatic saving. Cable deployment, certified terminations, essential power, actual soils, rescue/evacuation access and assisted passenger operation remain physical gates.

City manufacturing records carry material quantities, process routes, equipment needs and construction approaches. Candidate processes include controlled GFRP infusion/RTM and bonding, precast mould/cage/prestress/cure work, reaction-iron and carrier fabrication, station coil winding, cartridge assembly, wiring and tests. The production model takes the minimum of factory, foundation, pier, delivery, erection and final survey capacities. Its 56.25 route-m/planned-day target is not measured output. Demould, connection strength, crane-at-radius, ground and final QA gates remain unexecuted.

Prefer verified British manufacture when considering UK machinery. A UK supplier or integrator does not establish manufacturing origin. Complete local content remains unknown until origin-traced bills and configured offers exist.

Evacuation and publication

The preferred research layout has one beam per direction without a continuous external walkway. Necessary magnetic faces, supports and connections remain. Vertical spiral chute research covers variable height, anchorage, landings, assistance, fire/weather, partial failure, rescue and driverless use. Walkway removal needs accepted route and onboard alternatives at every allowed stopping point.

The moving-train video uses calculated six-car positions and beam states with declared visual amplification. Other CAD videos are final-position visibility sequences; none establishes collision-free erection, crane performance or chute deployment. CAD integrity, PDE residuals, virtual controller ticks and consistent publication hashes each establish only their checked digital scope. Closure work and v0.5.2 development notes remain explicit.

osr-maglev-control

crates/osr-maglev-control/README.md

Portable fixed-zone ferrite-maglev study core, using Rust `no_std`, no heap and no external dependencies. It targets simulation and off-the-shelf SBC replay. Station and located-booster LSM demand is separate from onboard active PM support and guidance. Every computed output has `physical_output_authorised = false`; the I/O boundary records study outputs and supplies no current, gate-drive or PWM operation.

The existing `evaluate(previous, inputs, now_ns, params)` API remains a single-module integer current hypothesis. Its polarity, damping, envelope, time/replay, stop/reset and support-power tests remain compatible. Increasing an upward-attraction gap asks for increasing support. The controller cannot be reused unchanged for shaped-pole passive vertical support.

24-module force and moment study

`array::evaluate_array` adds coupled heave, roll, pitch, lateral displacement and yaw feedback and `array::allocate` distributes requested forces and moments over 24 support stations. This is one rigid car; the complete moving six-car Baghdad train and a validated six-degree-of-freedom suspension remain open. The array model has no longitudinal actuator; nonzero longitudinal demand appears as an unachieved residual. The separate station module supplies an offline LSM demand screen, without closing physical propulsion/braking, traction/support coupling, flexible vehicle/beam modes, finite-angle dynamics or contact behavior.

Coordinates use x forward, y left and z up, with the assumed car centre of mass as origin. Station positions and gaps are metres, force is newtons, moments are newton-metres, small angles are radians and timestamps are monotonic nanoseconds. Upward force is positive attraction. Lateral force is positive left. Each moment is the sum of `position cross force`. The explicit seed geometry is 12 paired stations at x = -5.5, -4.5, …, 5.5 m and y = ±0.88 m, z = 0 m; this matches the candidate 12 m magnetic strips. Actual centre of mass, attachment heights, lever arms, tolerances and dynamic mass properties need physical closure.

Each station has a bounded upward channel and a signed lateral channel. The illustrative limits are 0–20 kN upward, ±4 kN lateral and 2 MN/s slew per channel. These numbers are hypothetical force envelopes, not measured magnetic maps, current/voltage/thermal limits or proved simultaneous lift/guidance capability. A real PM-biased module can retain force after drive loss: a failed station takes an explicit specified residual force. Zero in the example is an ideal failure assumption. Unknown failed-channel forces require a separate bounded uncertainty and protection model before hardware use.

The allocator uses at most 256 bounded coordinate sweeps to minimise squared force residuals and moment residuals scaled by a declared 6 m lever arm. It reports the independently reconstructed resultant, residual wrench, constraint limits and feasibility tolerances. A solver iteration count establishes no physical deadline. Infeasible requests remain visible; no six-axis success is inferred from solver completion. Slew limits apply against a prior force frame; without one the first study allocation has no force history and applies only static limits. This is not a physical startup or touchdown procedure.

The mode estimator fits a small-angle rigid plane to 24 dual-gap samples and fits lateral displacement/yaw to 24 dual lateral samples. Every individual gap and lateral sample is checked before averaging; disagreement, nonfinite values and degenerate geometry fail. This fit is not a flexible-body observer. Sensor channels share one supplied acquisition timestamp; per-channel skew, physical independence, common-cause failure and calibration are not established. Derivatives use acquisition time. Host time/sequence checks reject stale, future, replayed, regressing and late frames.

A stop request preserves fresh bounded support. A station failure, active bound or infeasible allocation latches a traction stop while available ideal support remains active. Missing valid sensing or power requests external local fallback, never frozen study forces. A prior fault inhibits traction until a fresh healthy frame, reported stopped/secured conditions and reset coincide. Capture, fallback, independent protection energy and secure-condition verification need separate engineering; they are not supplied by these Boolean reports.

Reproducible offline checks

Run:

```
cargo test -p osr-maglev-control
cargo clippy -p osr-maglev-control --all-targets -- -D warnings
cargo run -p osr-maglev-control --example sbc_replay
cargo run -p osr-maglev-control --example multimodule_replay
```

The array replay integrates ten seconds of virtual 1 ms frames, prints CSV at 500 ms intervals and asserts feasibility and a final small five-mode error. Its 27,750 kg AW2 mass is a retained numerical comparison: 18,750 kg tare plus 120 × 75 kg passengers. The current station-powered design instead targets 14,200 kg tare; this retained control case is not the new

vehicle mass closure. Gains, diagonal inertias, force limits and instantaneous actuator response are synthetic. It starts with coupled displacement/tilt, commands a stop at 0.5 s and loses station 0 at 1 s under a specified zero-residual failure hypothesis. It keeps fresh support and the stop latch active. This trajectory is a numerical example, not a closed-loop stability or ride-quality proof.

`ideal::IdealPlant` integrates a constant wrench exactly within each small-angle step. Tests independently check uniform equilibrium (272,227.5 N weight), moment signs, force/moment reconstruction, five-component demand, failed-force redistribution, capacity and slew infeasibility, all-failed residuals, sensor and time rejection, support-preserving stops and secure reset. Mechanical interface power agrees with resultant wrench power, and one-step work agrees with change in kinetic plus gravitational energy. These conservation checks validate the ideal mechanics arithmetic, not actual magnet energy or electrical losses.

`io::StudyIo`, `host_tick` and `ReplayIo` define an acquisition/record boundary without owning an OS clock or hardware. Recording/acquisition failures return to the host; the host must engineer its local fallback. Repeated single-frame stub reads are rejected as replay. No scheduler, GPIO, ADC or network transport is validated, and this example measures neither SBC timing nor I/O latency.

Physical gates remain open: calibrated nonlinear magnetic force/loss maps over gap/current/temperature and shared flux, identified electrical/thermal and mechanical dynamics, actuator failure residuals, independent sensors/protection, measured end-to-end target timing, multi-module rigs, flexible guideway and loaded vehicle tests, braking/capture and train/route acceptance. No operational or manufacturing authority is inherited from the numerical study.

Flexible guideway numerical demonstrator

`coupled_bridge` connects this exact array core to the separately labelled Python beam demonstrator in `engineering/ferrite_maglev/coupled.py`. From the repository root, run:

```
cargo build -p osr-maglev-control --release --example coupled_bridge
.venv/bin/python -m engineering.ferrite_maglev.coupled --duration-s 3
.venv/bin/python -m pytest -q engineering/analysis/tests/test_ferrite_maglev_coupled.py
```

The Python host sends all 24 local relative gaps each virtual millisecond. Rust returns bounded support forces, which load a 25 m simply supported Euler–Bernoulli beam; its local displacement feeds the next sensor frame. This is two-way offline coupling, with no physical output. The beam assumes the current configured section and native quantity-derived line mass, with 2% modal damping, retaining six vertical modes of a 50-element consistent-mass model. Independent point-load closed forms check static response; analytical pinned-beam frequencies and virtual-work/undamped-energy checks control the numerical implementation.

The model can show global beam displacement exceeding 8 mm while the car follows that displacement and the relative gaps remain inside the example's study envelope. This establishes neither a gap requirement nor structural acceptance. It omits dead-load camber, torsion/corbel flexibility, moving/train loads, flexible car structure, soil, joints, fatigue, real magnetic/electrical dynamics and independent protection. The line protocol accepts at most 20,000 frames, 8 KiB per frame, and measures no hardware timing. The three-mode car motion and vertical beam coupling do not provide a six-degree-of-freedom stability proof.

Offline evacuation sequencing

`egress::evaluate` adds allocation-free doorway sequencing with an exact declared candidate/door context, fresh monotonic frames, explicit readiness/assistance prerequisites, latched contradiction/context/deadline faults, and secured/stowed reset. It reports study traction inhibition and continued-support obligations; it supplies no door, restraint, chute or support output adapter. Input assertions are unauthenticated study facts, not verified independent sensor or safety evidence.

`cargo run -p osr-maglev-control --release --example egress_replay -- REVISION PATH_CLEAR HEIGHT_COVERED` prints a bounded nine-frame synthetic sequence. The linked Python study supplies its candidate tag and calculated hypothetical path/height gates, and retains negative controls. Loss of prerequisites after deployment begins holds the sequence; no automatic retraction is commanded. The egress/support composition test preserves fresh support control while inhibiting traction. Physical timing, independent protection, accessible passenger evacuation and railway approval remain open.

Station launch and receiving study

`station` evaluates bounded offline launch, coast, receive and secured modes. It checks exact configuration/zone identity, fresh monotonic frames, phase uncertainty, reservation assertions, magnetic-band overlap and assumed current, voltage and thermal limits. Coast demands zero station thrust. Receiving braking requires the declared brake resistor to be ready. Receiver loss after commitment requires independent stopping/recovery; clearing a Boolean drive demand does not stop an already coasting train.

```
cargo run --locked -p osr-maglev-control --example station_replay
./osr design protocol
```

The scalar force/current relation, winding R/L/flux and voltage/loss calculations are explicit hypotheses. They are not measured motor maps, three-phase hardware commutation, shared support/thrust qualification or a six-car electromagnetic simulation. An external `launch_committed` assertion is not an authenticated interlocking or detector record. Persistent corridor/berth ownership is exercised separately in `osr-interlocking`; permission expiry cannot release a committed train, and full-consist clearance is required. No physical output adapter or railway operating authority is supplied.

The longer-corridor study (`./osr design corridor`) separately evaluates urban 2 km and suburban 5 km profiles, finite station/booster overlap and electrical rating alternatives. A booster is a nonpassenger powered asset; junctions are passenger transfers between independent fixed lines. The 1 km station replay remains a component benchmark. Its passing result cannot clear longer-profile voltage, service-energy, recovery, section-handover or exclusive-reservation capacity holds.

Adaptive research tuning

`./osr design iterate` varies bounded proportional and damping gains alongside vehicle mass allocations and guidebeam sections/spans. The bridge accepts mass and gain scales as optional command arguments; nonfinite or out-of-bound scales are rejected before processing frames. Exact revisions face separate confirmation scenarios and replay/excess-gap negative controls. This offline tuning establishes neither measured magnetic response nor stability over the full operating envelope.

OpenSourceRail architecture

docs/ARCHITECTURE.md

OpenSourceRail v0.5.2 development uses fixed-zone ferrite maglev. Fixed station linear synchronous motors launch and receive each train; located midpoint accelerators are alternatives for longer suburban legs. Coasting sections outside these finite powered zones have passive magnetic reaction iron. Onboard ferrite PM-biased active support and guidance remain powered independently, alongside passenger services, protection and emergency equipment.

The objective is broad city access with compact elevated guidebeams, fewer moving running-gear parts and practical local manufacture. A 99% elevated-mainline planning target still depends on route rights, utilities, ground, curvature, station access, evacuation and recovery.

Configuration and interfaces

Baghdad uses six 17 m bodies. The 104.5 m coupled footprint is an assumption until couplers and gangways are resolved. The central study targets 14.2 t tare/car, including a 5.2 t equipped GFRP sandwich body, 3 t ferrite, 30 kWh services storage and 1.8 t growth reserve. Four doors per car are the body-study basis. No complete weighed body, load-bearing joint set or qualified magnetic circuit establishes these mass targets.

The ordinary civil candidate is a 25 m, 0.80 × 2.50 m box with 100 mm webs, 150 mm flanges and end infills. Hollow columns and caps are explicit quantity studies. Shallow, single-shaft and pile-group foundations remain alternative families, selected only with ground and structural evidence. Reaction iron, anchors, bearings, joints, rail survey and all construction stages remain in scope.

Layer	Controlled definition	Implemented evidence and boundary
Service and city	Geography, demand, formation, runtime, headway and station spacing	Urban minimum target 1.5 km and mean target 2 km; explicitly classified suburban legs 4–5 km with booster alternatives; short interchange conflicts remain visible
Mechanical	Structural sandwich panels, four doors, ferrite cartridges, support/guidance and services	Quantity takeoffs and partial solids; complete hardpoints, joints, mass/CG/inertia and qualification open
Fixed-zone propulsion	Station/booster LSM cassettes/groups, phase sensing, bridges, DC supply, receiving braking and charging	Separate longer-leg jerk/overlap/rating/service/protection comparisons; measured force/current/voltage/thermal maps unqualified
Civil	Guidebeams, supports, foundations, stations and interfaces	Exact section/end-infill quantities, gross elastic screens and foundation sensitivities; no accepted strength or soil capacity
Software	Independent support control and persistent launch/corridor/berth ownership	Offline Rust tests/replays; no qualified sensing, physical actuator adapter or measured hardware timing
Industrialisation	Moulds, cure/release, crews, transport, cranes, joints and survey	Conditional bottleneck capacity; no measured production or supplier-price evidence
Acceptance	Exact configuration, source, unit/serial identity, tests and reviewers	Source-bound receipts; engineering/manufacturing/physical acceptance remains open

Infrastructure policy, configuration, operating study and structure study control the interfaces. City `city.json` inputs feed the same current compiler. Junctions are passenger transfers between independent fixed lines, with no mainline train-routing switch. Straight alignments are the design objective; retained mapped routes and unselected straight-chord alternatives must keep their site and cost evidence separate.

Launch, receiving and recovery

A launch reserves the interstation corridor, receiving berth and protective boundary for one train. Permission expiry does not release committed occupancy. Release requires complete-consist clearance evidence; lost communications or a timer cannot establish that clearance. The receiver must remain able to stop a train after departure. Independent stopping/holding, receiver faults, low-service energy and stranded-train recovery require their own engineered designs.

Coasting feasibility depends on grade, drag, wind, launch speed and arrival margin. An unpowered span cannot restart a stationary train from a distant station. Terminal return geometry, depot movement and rescue equipment remain explicit requirements. The nominal trajectory does not close these obligations.

The 1 km/90 s authored and 1 km/88 s developed runs remain explicit benchmarks. They do not verify the new 2 km/5 km operating design. Longer profiles compare converter ratings, charger power, dwell and larger services packs with their mass and cost changes. Whole-leg exclusive reservations impose a capacity constraint; a passing motion calculation cannot clear it. Station and booster power/charging are separately priced targets from the incomplete civil subtotal. Complete network economics must include both, plus stations, fleet, recovery, land, qualification, factory capital and financing.

Evidence boundary

Ring-field controls and block orientation seeds do not demonstrate railway lift, thrust or magnet-mass gains. Gross beam sag is distinct from relative magnetic-gap error. The stationary single-car demonstrator is retained separately from the moving six-car Rust/beam/coupler model. Both use ideal force channels. Longer operational profiles need their own moving confirmation; flexible bodies, torsion, joints, soil and calibrated electromagnetic/thermal response remain open.

Native CAD covers partial assemblies. Visibility videos show final-position parts and an illustrative sequence; they do not establish collision-free motion, erection or chute deployment. Continuous-walkway removal remains conditional on accepted onboard and route evacuation, including assisted access and safe landings at every permitted stop. All release flags remain false. Accepted engineering and physical releases remain zero.

Roadmap

docs/ROADMAP.md

OpenSourceRail v0.5.2 development coordinates fixed station/booster-zone ferrite maglev across mechanical, civil, software, manufacturing and city planning. Station LSMs launch and receive the train; longer suburban legs compare located midpoint accelerators. Independent onboard support and passenger services remain active during the unpowered coast. Junction stations transfer passengers between fixed lines. Urban spacing targets 1.5 km minimum and approximately 2 km average, with explicitly classified suburban 4–5 km alternatives. Implemented work and the current report identify the tested digital scope. Accepted engineering and physical releases remain zero.

Primary maglev closure queue

Package	Evidence required for closure
Requirements and configuration	Approved six-car loads, service/grade envelope, door layout, tolerances and quantitative acceptance criteria
GFRP body	Sized orthotropic panels, longitudinal members, magnetic hardpoints, joints, openings and coupler/crash paths; fire, heat, fatigue, repair and weighed equipment closure
Magnetic assemblies	Complete shared or separately priced support/guidance/thrust circuits, cartridge retention, capture, sensors and tolerances
Magnetic force and loss	Converged finite-geometry/steel/coil maps over gap, phase, current and temperature, followed by rig correlation; no automatic ring-paper gain credit
Station and booster propulsion	Located fixed-group/section topology, phase registration, winding R/L/flux, current/voltage/thermal limits, receiving brakes, fault response, utilities and actual purchasing quantities
Service energy and charging	Verified hot/normal duty, battery life/reserve, complete dwell charging, power-loss support and recovery mission
Civil kit	Prestress/rebar/cover/end blocks, bearings, rail anchors, joints, hollow support connections, ground-specific foundations and every handling/construction stage
Launch and protection	Atomic corridor/receiving-berth reservation, persistent committed ownership, proved tail clearance, independent receiving readiness and stopping/holding under faults
Coupled dynamics	Moving six-car loads, flexible body, lateral/torsional response, curves, joints and soil with calibrated magnetic/electrical/thermal dynamics
Hardware	Isolated I/O, current drives, protection/watchdogs and measured end-to-end timing on selected SBC/MCU equipment
City delivery	Grade/coasting arrival margins, surveyed routes/utilities, terminal return geometry, depot/recovery access and evacuation at every stop location
Factory and erection	Measured mould/cure cycles, strength releases, delivery/crane-at-radius plans, joint completion, survey acceptance and bottleneck output
Economics and origin	Complete origin-traced MBOM, configured supplier offers, station/civil/fleet scope, capital/tooling, maintenance, financing and uncertainty
Demonstrator	Manufactured serials and exact revisions, independent inspections, measured system operation and physical acceptance

The implemented station study adds jerk-limited launch/coast/arrival calculations, magnetic-band overlap, receiving-brake duty and headway/power/cost sensitivities. The structure study adds quantity-derived GFRP cases, orientation seeds, five beam families, support/foundation screens and a two-shift bottleneck model. These are calculated candidates and production targets; they do not supply the missing structural, magnetic or measured factory evidence.

The adaptive designer uses preceding results to propose vehicle-mass, beam/span and actual Rust-gain variants. It preserves ancestry/traces and uses separate confirmation cases. Its original search plant is a stationary single rigid car; the moving six-car plant and longer-corridor confirmations are separate evidence. All currently use ideal magnetic force channels. A preferred candidate remains a bounded research result, without automatic promotion to accepted geometry or city criteria.

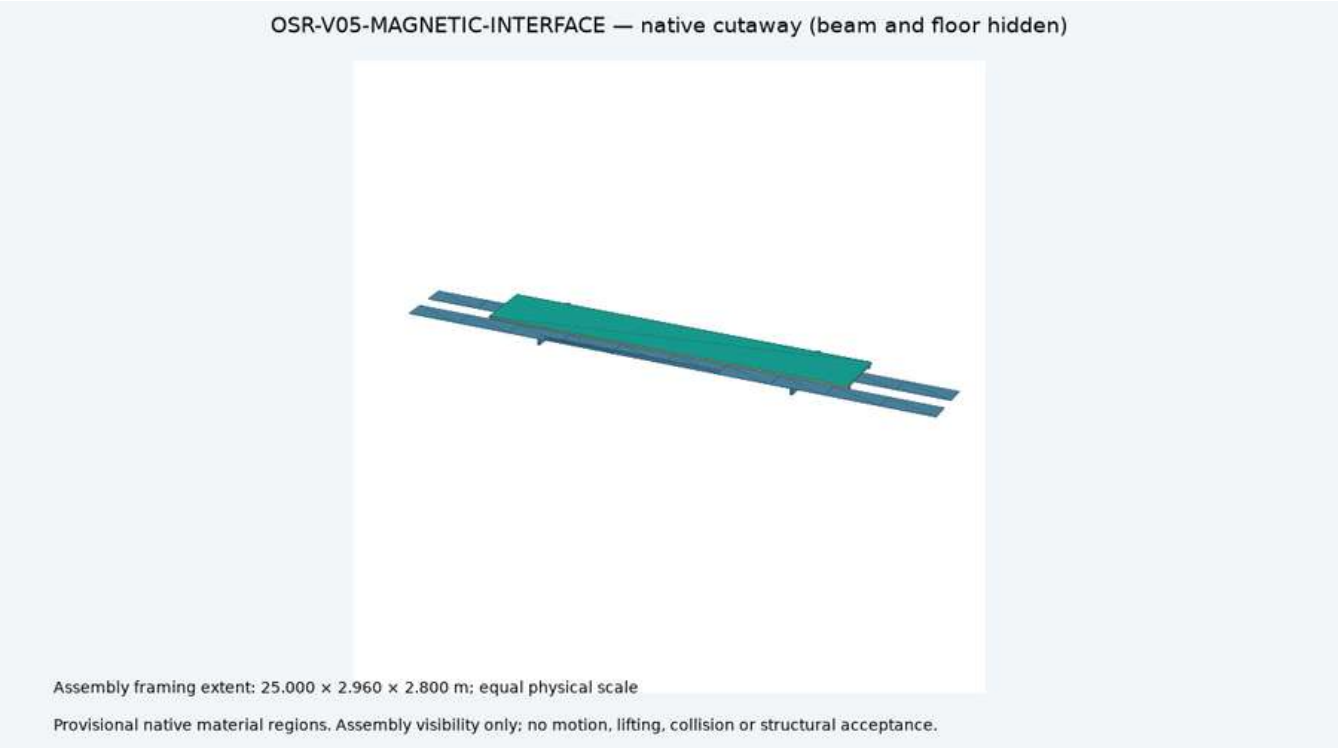
The dependency runner rebuilds stale implemented studies and holds downstream outputs after failures. Completing every automated check does not complete the engineering queue. Published v0.5.1 remains a distinct preserved release; this revision is documented in v0.5.2 development notes.

Native geometry review



Partial six-car Baghdad station, magnetic guidebeams and island lift; acceptance remains open.

Partial six-car Baghdad station, magnetic guidebeams and island lift; acceptance remains open.



Partial fixed magnetic interface; not a complete vehicle.

Partial fixed magnetic interface; not a complete vehicle.



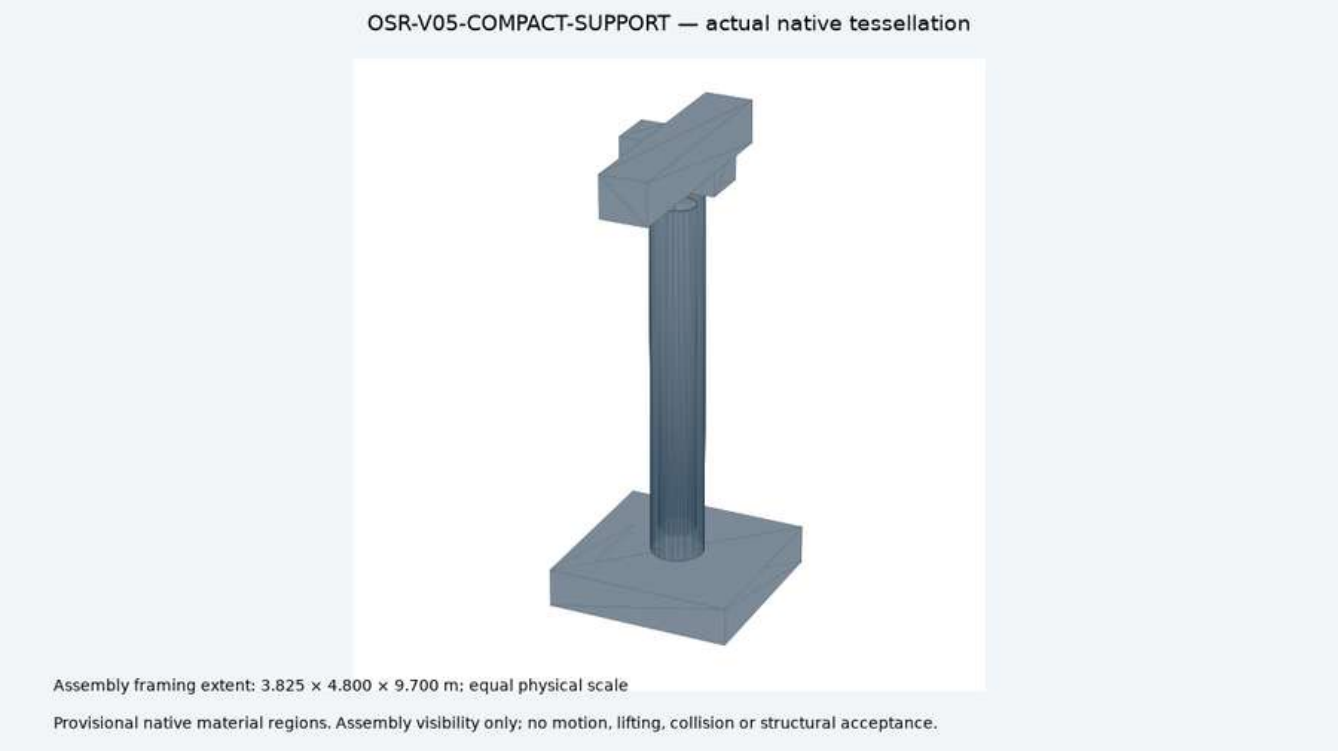
Beam-only civil variant. Supports are envelopes; evacuation acceptance remains open.

Beam-only civil variant. Supports are envelopes; evacuation acceptance remains open.



Partial evacuation entry frame and empty cassette. No deployed or stowed mechanism is qualified.

Partial evacuation entry frame and empty cassette. No deployed or stowed mechanism is qualified.



Compact pier and foundation envelope. Site capacity and structural acceptance remain open.

Compact pier and foundation envelope. Site capacity and structural acceptance remain open.

Primary maglev city directory

Separate planning packages. Complete programme costs, local content, land and physical acceptance remain open for every city.

City	Provisional trainsets	Cars	Open gates
Baghdad	451	2706	12
Benguela	87	261	12
Huambo	79	237	12
Lobito	57	171	12
Luanda	436	2616	12
Lubango	75	225	12
Malanje	24	72	12
Namibe	66	132	12
Soyo	51	102	12
Uige	16	48	12
Bukavu	89	267	12
Goma	72	216	12
Kananga	45	180	12
Kinshasa	600	3600	12
Kisangani	47	188	12
Lubumbashi	168	672	12
Mbuji Mayi	124	496	12
Eldoret	94	282	12
Garissa	49	98	12
Kakamega	61	122	12
Kisii	45	90	12
Kisumu	74	222	12
Kitale	63	126	12
Machakos	49	98	12
Malindi	49	98	12
Meru Ke	53	106	12
Mombasa	254	1016	12
Nairobi	540	3240	12
Naivasha	60	120	12
Nakuru	81	243	12
Nyeri	55	110	12
Thika	95	285	12
Antananarivo	347	2082	12
Beira	66	198	12
Chimoio	50	150	12
Lichinga	26	52	12
Maputo	204	816	12
Nacala	68	136	12
Nampula	75	225	12
Pemba Mz	54	108	12
Quelimane	15	45	12
Tete	69	207	12
Xai Xai	41	82	12
Huye	67	134	12
Kigali	186	744	12
Rubavu	62	124	12
Mogadishu	143	572	12
Waw	35	70	12
Arusha	80	240	12
Dar Es Salaam	523	3138	12
Dodoma	75	225	12
Iringa	51	102	12
Kigoma	49	98	12
Mbeya	69	207	12
Morogoro	68	204	12
Moshi	53	106	12
Mwanza	209	836	12

City	Provisional trainsets	Cars	Open gates
Shinyanga	55	110	12
Songea	21	42	12
Sumbawanga	30	60	12
Tabora	45	90	12
Tanga	70	210	12
Zanzibar City	81	243	12
Arua	56	112	12
Entebbe	65	130	12
Fort Portal	57	114	12
Gulu	96	288	12
Hoima	51	102	12
Jinja	76	152	12
Kampala	242	968	12
Lira	66	132	12
Masaka	52	104	12
Mbale	66	132	12
Mbarara	76	228	12
Soroti	22	44	12
Lusaka	284	1704	12
Lyon	279	1116	12
La Paz	232	928	12
Cuenca	93	279	12
San Salvador	266	1064	12
Agadir	112	336	12
Beni Mellal	44	88	12
Fez	137	548	12
Kenitra	81	243	12
Khouribga	29	58	12
Marrakech	221	884	12
Meknes	65	195	12
Nador	57	114	12
Oujda	50	150	12
Safi	59	177	12
Tangier	217	868	12
Tetouan	79	237	12
El Obeid	76	228	12
Kassala	47	141	12
Khartoum	435	2610	12
Nyala	67	201	12
Omdurman	290	1160	12
Port Sudan	54	162	12
Tunis	247	988	12
Bloemfontein	111	333	12
Durban	417	2502	12
East London Za	85	255	12
Nelspruit	54	108	12
Polokwane	78	234	12
Herat	59	177	12
Jalalabad Af	67	201	12
Kabul	254	1524	12
Kandahar	76	228	12
Mazar E Sharif	90	270	12
Barisal	95	285	12
Chittagong	354	2124	12
Comilla	95	285	12
Gazipur	269	1076	12
Khulna	226	904	12
Mymensingh	72	216	12
Narayanganj	102	306	12
Rajshahi	62	186	12
Rangpur	73	219	12
Sylhet	76	228	12
Agra	144	576	12

City	Provisional trainsets	Cars	Open gates
Bhopal	180	720	12
Coimbatore	314	1884	12
Indore	310	1860	12
Jodhpur	142	568	12
Kanpur	304	1824	12
Lucknow	309	1854	12
Madurai	205	820	12
Meerut	126	504	12
Patna	250	1000	12
Raipur	136	544	12
Rajkot	114	456	12
Ranchi	210	840	12
Vadodara	149	596	12
Varanasi	196	784	12
Vijayawada	184	736	12
Visakhapatnam	224	896	12
Biratnagar	54	108	12
Kathmandu	235	940	12
Pokhara	109	327	12
Bahawalpur	69	207	12
Faisalabad	181	1086	12
Gujranwala	202	808	12
Hyderabad Pk	216	864	12
Karachi	508	3048	12
Larkana	48	144	12
Multan	142	568	12
Peshawar	193	772	12
Quetta	124	496	12
Rahim Yar Khan	84	252	12
Sheikhupura	23	69	12
Sialkot	77	231	12
Sukkur	65	195	12
Colombo	345	2070	12
Galle	94	282	12
Jaffna	86	258	12
Kandy	97	291	12
Phnom Penh	280	1120	12
Bandung	243	972	12
Surabaya	284	1704	12
Vientiane	100	300	12
Mandalay	237	948	12
Yangon	474	2844	12
Davao	268	1072	12
Ouagadougou	227	908	12
Bafoussam	89	267	12
Bamenda	68	204	12
Bertoua	32	96	12
Douala	202	1212	12
Edea	12	24	12
Garoua	54	162	12
Kumba	56	168	12
Maroua	74	222	12
Ngaoundere	41	123	12
Yaounde	224	1344	12
Conakry	105	420	12
Bamako	291	1164	12
Niamey	193	772	12
Aba Ng	51	153	12
Benin City	149	596	12
Ibadan	152	912	12
Ilorin	74	222	12
Jos	79	237	12
Kano	429	2574	12

City	Provisional trainsets	Cars	Open gates
Maiduguri	200	800	12
Onitsha	210	840	12
Port Harcourt	200	800	12
Uyo	55	165	12
Dakar	250	1500	12
Arish	23	46	12
Asyut	76	228	12
Beni Suef	56	168	12
Damanhur	77	231	12
Damietta	102	306	12
Fayoum	85	255	12
Hurghada	51	102	12
Ismailia	71	213	12
Kafr El Sheikh	53	106	12
Luxor	81	243	12
Mahalla	61	183	12
Mansoura Eg	72	216	12
Minya	68	204	12
Port Said	49	147	12
Qena	81	243	12
Sohag	58	174	12
Suez	85	255	12
Tanta	97	291	12
Zagazig	57	171	12
Amarah	70	210	12
Baqubah	94	282	12
Basra	304	1824	12
Diwaniyah	78	234	12
Duhok	80	240	12
Erbil	171	684	12
Fallujah	77	231	12
Hillah	88	264	12
Karbala	193	772	12
Kirkuk	168	672	12
Kut	69	207	12
Mosul	221	884	12
Najaf	198	792	12
Nasiriyah	76	228	12
Ramadi	74	222	12
Samawah	77	231	12
Sulaymaniyah	124	496	12
Amman	372	2232	12
Aqaba	59	118	12
Irbid	74	222	12
Zarqa	98	294	12
Beirut	182	728	12
Sidon	51	102	12
Tripoli Lb	69	207	12
Gaza City	51	153	12
Hebron	87	261	12
Nablus	96	288	12
Abha	95	285	12
Al Kharj	88	264	12
Buraidah	97	291	12
Dammam	285	1140	12
Hail	89	267	12
Hofuf	100	300	12
Jeddah	365	2190	12
Jizan	78	234	12
Khamis Mushait	115	345	12
Mecca	234	936	12
Medina	196	784	12
Najran	83	249	12

City	Provisional trainsets	Cars	Open gates
Tabuk	87	261	12
Taif	85	255	12
Aleppo	224	896	12
Damascus	238	952	12
Deir Ez Zor	84	252	12
Hama	84	252	12
Homs	61	183	12
Idlib	54	108	12
Latakia	69	207	12
Raqqa	59	177	12
Tartus	46	92	12
Aden	74	222	12
Dhamar	58	116	12
Hodeidah	47	141	12
Ibb	79	237	12
Lahij	48	96	12
Mukalla	88	264	12
Sanaa	292	1752	12
Sayun	41	82	12
Taiz	78	234	12